

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IX

Roll No: T001	Name: Soham Acharekar
Class: TYIT	Batch: 1
Date of Assignment: 1/9/26	Date/Time of Submission: 1/9/26

Part B – Program

Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

Student ID	Full Name	Dept Info	Math	Physics	Chemistry
S101	Aarav Sharma	CSE-01	85	0.900000	88
S102	priya VERMA	ECE-02	78	0.750000	80
S103	Rohan Gupta	CSE-01	78	0.820000	85
S104	Sneha Patel	IT-03	92	0.880000	90
S105	aditya Rao	MECH-04	45	0.700000	50
S106	Ananya Singh	CSE-01	88	0.910000	85
S107	Rahul Nair	ECE-02	95	0.700000	75
S108	Kavya Reddy	IT-03	95	0.940000	92

Attendance_Clean

`FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

The calculation is valid.

2 Dependencies

Apply

OK

Chemistry_Clean

`IFNULL([Chemistry], 0)`

The calculation is valid.

10 Dependencies

Apply

OK

Maths_Clean

X

Physics_Clean

X

IFNULL([Math], 0)

IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)

The calculation is valid.

10 Dependencies ▾

Apply

OK

The calculation is valid.

10 Dependencies ▾

Apply

OK

b) Convert wide data into tall format

Attendance ▾	Sem1_Scores.csv	Attendance ▾	Abc ▾	# ▾
Pivot Field Na ▾	Pivot	Pivot Field Na...	Pivot	Pivot
Pivot Field Val ▾	Pivot	Pivot Field Val...	Subject	Score
Physics_Cle ! ▾	Calculation	Calculation_53...	Chemistry	88.0000
			Math	85.0000
			Physics	0.9000
			Programming	95.0000
			Chemistry	80.0000
			Math	null
			Physics	0.7500
			Programming	85.0000

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Attendance ▾	Sem1_Scores.csv	Attendance
Abc	Subject ▾	Pivot	Pivot Field Na...
#	Score ▾	Pivot	Pivot Field Val...
=#	Physics_Cle ! ▾	Calculation	Calculation_53...

Abc ▾	# ▾	=# ▾	=# ▾
Pivot	Pivot	Calculation	Calculation
Subject	Score	Physics_Clean !	Attendance_Clear
Chemistry	88.0000	null	85.0000
Math	85.0000	null	85.0000
Physics	0.9000	null	85.0000
Programming	95.0000	null	85.0000
Chemistry	80.0000	null	90.0000
Math	null	null	90.0000
Physics	0.7500	null	90.0000
Programming	85.0000	null	90.0000

c) Perform union of multiple files

Sem1_Scores.csv+

X

Semester

X

Specific (manual)

Wildcard (automatic)

Connection: Sem1_Scores

Sem1_Scores.csv X

Sem2_Scores.csv X

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"

ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"

ELSE "Other"

END

Tables in union: 2

Apply

OK

The calculation is valid.

3 Dependencies ▾

Apply

OK

d) Execute cross-database joins

Left Join Configuration:

Sem1_Scores.csv	Operator	Department_Master.csv
Abc Dept Info	=	Abc Dept Code

Right Join Configuration:

Sem1_Scores.csv	Operator	Student_Activities.csv
Abc Student ID	=	Abc Student ID (Student ...)

e) Restructure data for better visualization

Table View:

Student ID	Full Name	Semester	Performance_Band	ATTR(Attendance_Clean)	SUM(Overall_Percentage)
S101	Aarav Sharma	67.2%	69.2%		
S102	Priya Verma	41.4%	62.4%		
S103	Rohan Gupta	41.7%	63.2%		
S104	Sneha Patel	69.7%	72.0%		
S105	Aditya Rao	38.8%	43.1%		
S106	Ananya Singh	66.5%	68.5%		
S107	Rahul Nair	38.9%	58.7%		
S108	Kavya Reddy	71.5%	72.7%		
S109	Vikram Joshi	49.4%	51.9%		
S110	Pooja Iyer	63.0%	65.5%		

Calculation Field:

Total_Score

[Maths_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]

The calculation is valid. 9 Dependencies

Default Number Format [Total_Score] Dialog:

- Automatic
- Number (Standard)
- Number (Custom)
- Currency (Standard)
- Currency (Custom)
- Scientific
- Percentage**
- Custom

Percentage

Decimal places: 1

Buttons: Clear, OK, Cancel

f) Handle different levels of detail in datasets

Sem1_Scores

Search

Tables

Performance_Band

Section_ID

Semester

Student ID

Student_Display_Label

Table Name

Attendance

Attendance_Clean

Chemistry

Chemistry_Clean

Dept_Average_Perce...

Math

Maths_Clean

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tool...

Bench..

SUM(D..

SUM(O..

Columns

Semester

Rows

Dept Info

Student ID

Full Name

Sheet 2

			Semester	
Dept Info	Student I..	Full Name	Seme..	Seme..
CSE-01	S101	Aarav Sharma	67.2%	69.2%
	S103	Rohan Gupta	41.7%	63.2%
	S106	Ananya Singh	66.5%	68.5%
	S109	Vikram Joshi	49.4%	51.9%
ECE-02	S102	Priya Verma	41.4%	62.4%
	S107	Rahul Nair	38.9%	58.7%
	S110	Pooja Iyer	63.0%	65.5%
IT-03	S104	Sneha Patel	69.7%	72.0%
	S108	Kavya Reddy	71.5%	72.7%
MECH-04	S105	Aditya Rao	38.8%	43.1%

Benchmark_Status

Above Department B..

Below Department Be..

average_Percentage_LOD

{ FIXED [Dept Info] : AVG([Overall_Percentage]) }

The calculation is valid.

3 Dependencies

Apply

OK

g) Apply advanced data cleaning techniques

Sem1_Scores

Search

Tables

Performance_Band

Section_ID

Semester

Student ID

Student_Display_Label

Table Name

Attendance

Attendance_Clean

Chemistry

Chemistry_Clean

Dept_Average_Perce...

Math

Maths_Clean

Overall Percentage

Pages

Filters

Marks

Dept_Branch

Section_ID

Automatic

Color

Size

Text

Detail

Tooltip

Performance_...

SUM(Overall_P..

Columns

Semester

Rows

Student_Display_La..

Sheet 3

		Semester	
Student_Display_La..	Seme..	Seme..	
S101-Aarav Sharm..	67.2%	69.2%	
S102-Priya Verma (..	41.4%	62.4%	
S103-Rohan Gupta ..	41.7%	63.2%	
S104-Sneha Patel (..	69.7%	72.0%	
S105-Aditya Rao (..	38.8%	43.1%	
S106-Ananya Sing..	66.5%	68.5%	
S107-Rahul Nair (E..	38.9%	58.7%	
S108-Kavya Reddy ..	71.5%	72.7%	
S109-Vikram Joshi ..	49.4%	51.9%	
S110-Pooja Iyer (E..	63.0%	65.5%	

Performance_Band

Needs Academic Supp..

Second Class

First_Name

TRIM(SPLIT([Full Name], " ", 1))

The calculation is valid.

Apply

OK

Last_Name

TRIM(SPLIT([Full Name], " ", -1))

The calculation is valid.

Apply

OK